

ADVANCED COMPUTER NETWORKS
(MTITE101)

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Branch: **Information Technology**
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Practical 4:

Aim: To construct a dual-stack network topology in ns-3 using Python bindings to demonstrate concurrent IPv4 and IPv6 protocol execution, configure explicit IPv6 static routing tables, and analyze transition mechanisms.

2. Step-by-Step Implementation Guide

Step 1: Environment & Directory Setup

Open a terminal and navigate to the root directory of your ns-3 installation:

Bash
```
cd ~/ns-allinone-3.33/ns-3.33
```

Step 2: Script Creation

Create a new file named `dualstacktunnel.py` inside the `scratch/` directory.

Step 3: Script Execution

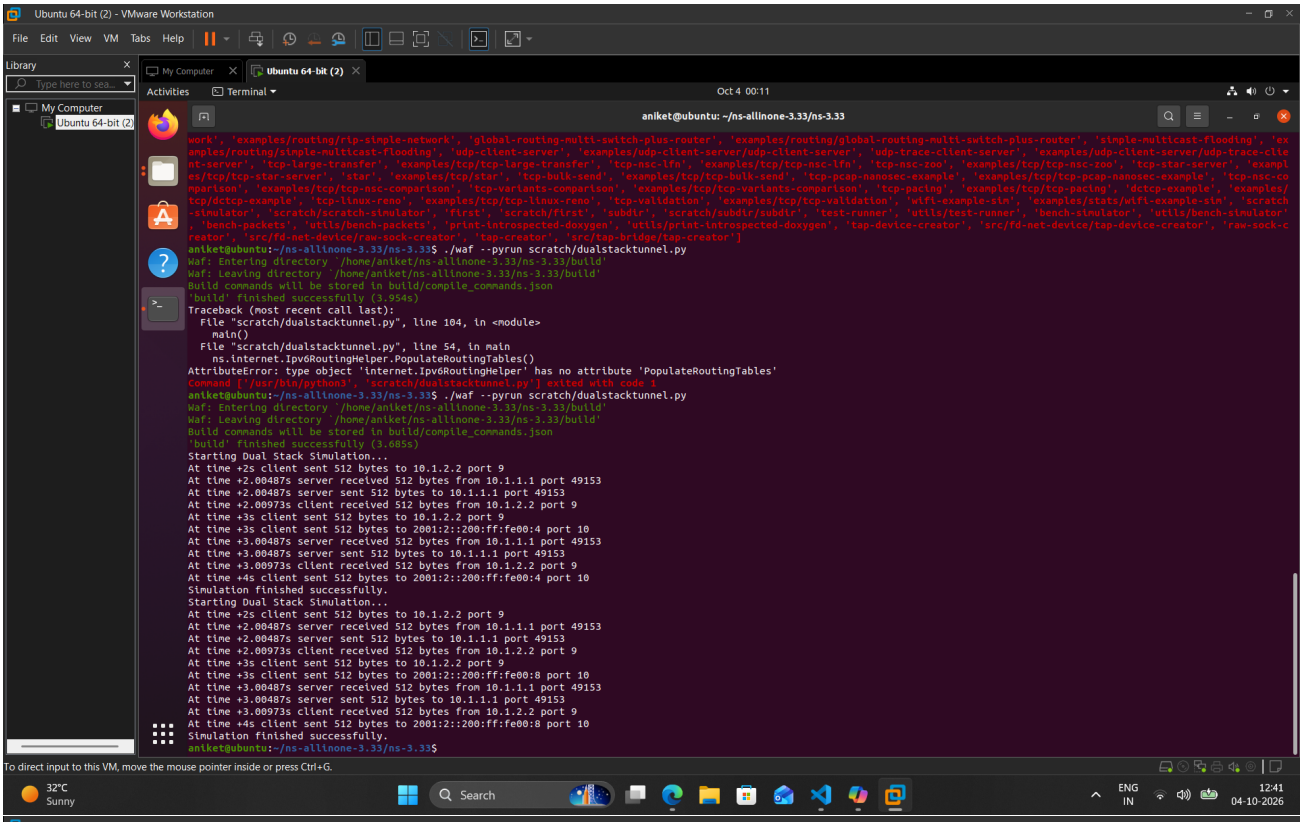
Execute the Python script through the ns-3 Waf tool:

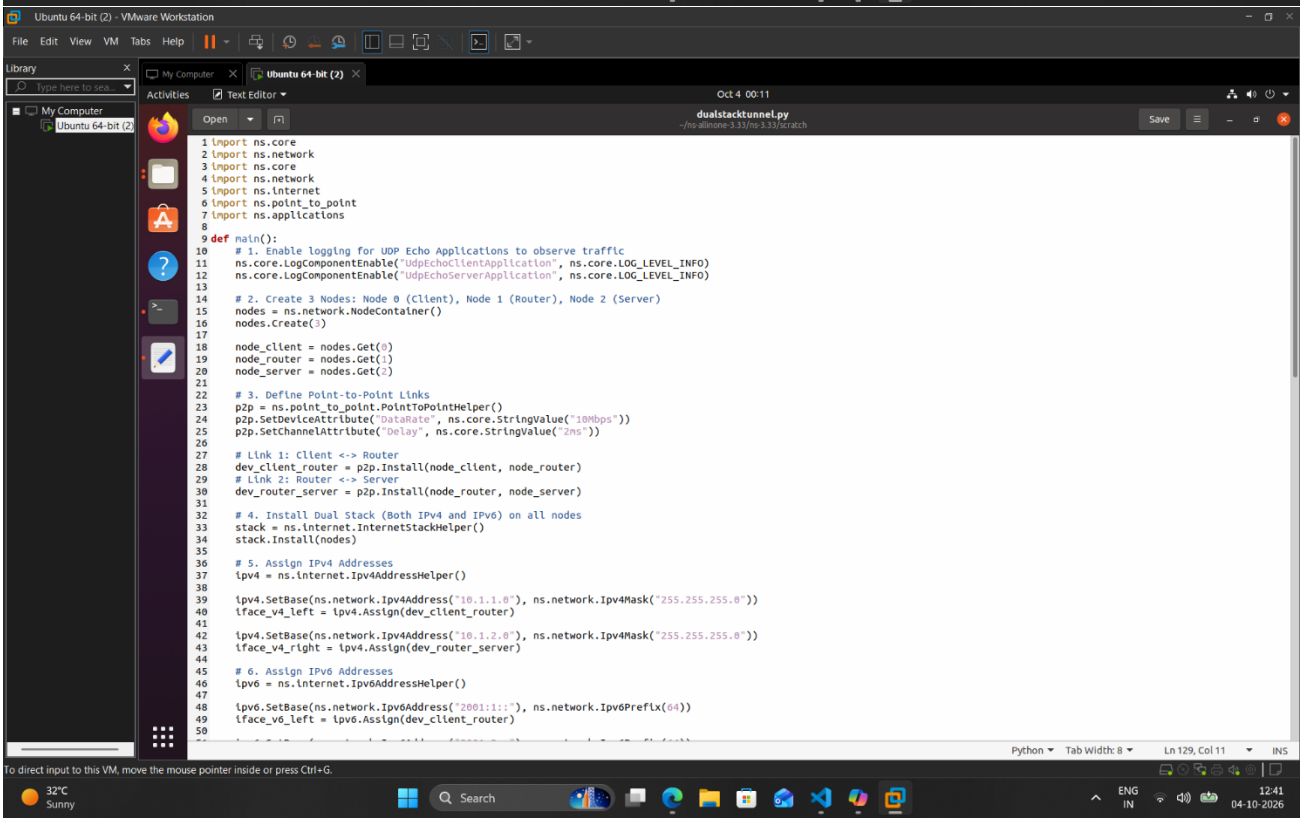
Bash
```
./waf --pyrun scratch/dualstacktunnel.py
```

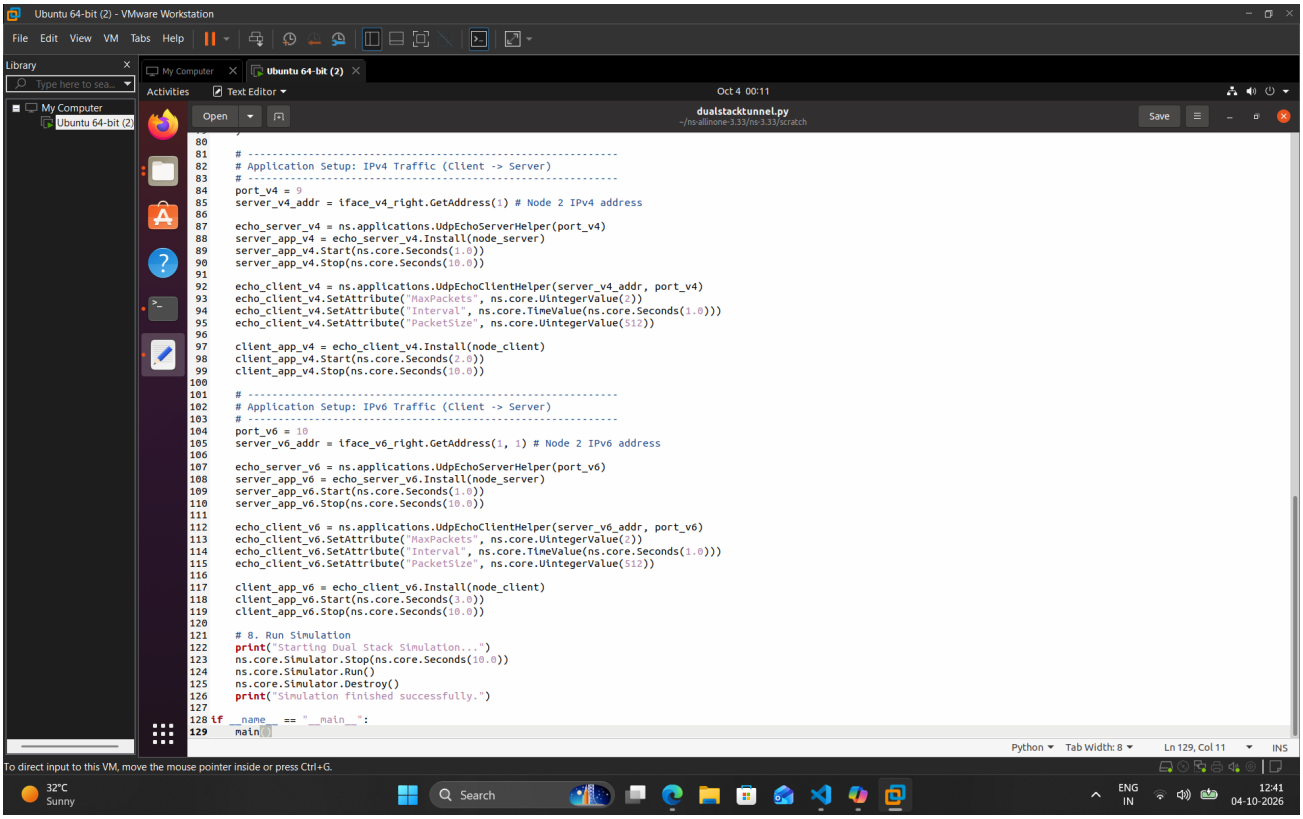
3. Dual-Stack & IPv6 Routing Mechanics

- **Dual-Stack Protocol Aggregation**: `InternetStackHelper` installs both `Ipv4L3Protocol` and `Ipv6L3Protocol` on each node. `Ipv4AddressHelper` and `Ipv6AddressHelper` bind IPv4 (`10.1.x.x`) and IPv6 (`2001:x::`) addresses concurrently to the same physical `NetDevice` interface.
- **Static Routing Requirement**: Unlike IPv4 (which provides `Ipv4GlobalRoutingHelper.PopulateRoutingTables()`), NS-3 lacks automatic global routing for IPv6. Routing across intermediate routers requires explicit static route configuration via `Ipv6StaticRoutingHelper.AddNetworkRouteTo()`.
- **Traffic Isolation**: The intermediate dual-stack router inspects incoming packet IP headers (vs) and routes each payload independently over the appropriate routing table without encapsulation overhead.

4. Execution Output & Terminal Logs:


Ubuntu 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Ubuntu 64-bit (2)
Activities
Terminal
Oct 4 00:11
aniket@ubuntu: ~/ns-allinone-3.33/ns-3.33
aniket@ubuntu:~/ns-allinone-3.33/ns-3.33$ ./waf --pyrun scratch/dualstacktunnel.py
Waf: Entering directory `/home/aniket/ns-allinone-3.33/ns-3.33/build'
Waf: Leaving directory `/home/aniket/ns-allinone-3.33/ns-3.33/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (3.954s)
Traceback (most recent call last):
File "scratch/dualstacktunnel.py", line 104, in <module>
main()
File "scratch/dualstacktunnel.py", line 54, in main
ns.internet.Ipv6RoutingHelper.PopulateRoutingTables()
AttributeError: type object 'internet.Ipv6RoutingHelper' has no attribute 'PopulateRoutingTables'
Command ['/usr/bin/python3', 'scratch/dualstacktunnel.py'] exited with code 1
aniket@ubuntu:~/ns-allinone-3.33/ns-3.33$ ./waf --pyrun scratch/dualstacktunnel.py
Waf: Entering directory `/home/aniket/ns-allinone-3.33/ns-3.33/build'
Waf: Leaving directory `/home/aniket/ns-allinone-3.33/ns-3.33/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (3.685s)
Starting Dual Stack Simulation...
At time +2s client sent 512 bytes to 10.1.2.2 port 9
At time +2.00487s server received 512 bytes from 10.1.1.1 port 49153
At time +2.00487s server sent 512 bytes to 10.1.1.1 port 49153
At time +2.00973s client received 512 bytes from 10.1.2.2 port 9
At time +3s client sent 512 bytes to 10.1.2.2 port 9
At time +3s client sent 512 bytes to 2001:2::200:ff:fe00:4 port 10
At time +3.00487s server received 512 bytes from 10.1.1.1 port 49153
At time +3.00487s server sent 512 bytes to 10.1.1.1 port 49153
At time +3.00973s client received 512 bytes from 10.1.2.2 port 9
At time +4s client sent 512 bytes to 2001:2::200:ff:fe00:4 port 10
Simulation finished successfully.
Starting Dual Stack Simulation...
At time +2s client sent 512 bytes to 10.1.2.2 port 9
At time +2.00487s server received 512 bytes from 10.1.1.1 port 49153
At time +2.00487s server sent 512 bytes to 10.1.1.1 port 49153
At time +2.00973s client received 512 bytes from 10.1.2.2 port 9
At time +3s client sent 512 bytes to 10.1.2.2 port 9
At time +3s client sent 512 bytes to 2001:2::200:ff:fe00:8 port 10
At time +3.00487s server received 512 bytes from 10.1.1.1 port 49153
At time +3.00487s server sent 512 bytes to 10.1.1.1 port 49153
At time +3.00973s client received 512 bytes from 10.1.2.2 port 9
At time +4s client sent 512 bytes to 2001:2::200:ff:fe00:8 port 10
Simulation finished successfully.
aniket@ubuntu:~/ns-allinone-3.33/ns-3.33$
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
32°C Sunny
Search
ENG IN
12:41 04-10-2026


Ubuntu 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Ubuntu 64-bit (2)
Activities
Text Editor
Oct 4 00:11
Open
dualstacktunnel.py
~/ns-allinone-3.33/ns-3.33/scratch
Save
import ns.core
import ns.network
import ns.core
import ns.network
import ns.internet
import ns.point_to_point
import ns.applications
def main():
1. Enable logging for UDP Echo Applications to observe traffic
ns.core.LogComponentEnable("UdpEchoClientApplication", ns.core.LOG_LEVEL_INFO)
ns.core.LogComponentEnable("UdpEchoServerApplication", ns.core.LOG_LEVEL_INFO)
2. Create 3 Nodes: Node 0 (Client), Node 1 (Router), Node 2 (Server)
nodes = ns.network.NodeContainer()
nodes.Create(3)
node_client = nodes.Get(0)
node_router = nodes.Get(1)
node_server = nodes.Get(2)
3. Define Point-to-Point Links
p2p = ns.point_to_point.PointToPointHelper()
p2p.SetDeviceAttribute("DataRate", ns.core.StringValue("10Mbps"))
p2p.SetChannelAttribute("Delay", ns.core.StringValue("2ms"))
Link 1: Client <-> Router
dev_client_router = p2p.Install(node_client, node_router)
Link 2: Router <-> Server
dev_router_server = p2p.Install(node_router, node_server)
4. Install Dual Stack (Both IPv4 and IPv6) on all nodes
stack = ns.internet.InternetStackHelper()
stack.Install(nodes)
5. Assign IPv4 Addresses
ipv4 = ns.internet.Ipv4AddressHelper()
ipv4.SetBase(ns.network.Ipv4Address("10.1.1.0"), ns.network.Ipv4Mask("255.255.255.0"))
iface_v4_left = ipv4.Assign(dev_client_router)
ipv4.SetBase(ns.network.Ipv4Address("10.1.2.0"), ns.network.Ipv4Mask("255.255.255.0"))
iface_v4_right = ipv4.Assign(dev_router_server)
6. Assign IPv6 Addresses
ipv6 = ns.internet.Ipv6AddressHelper()
ipv6.SetBase(ns.network.Ipv6Address("2001:1::"), ns.network.Ipv6Prefix(64))
iface_v6_left = ipv6.Assign(dev_client_router)
Python
Tab Width: 8
Ln 129, Col 11
INS
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
32°C Sunny
Search
ENG IN
12:41 04-10-2026



5. Explanation & Analysis

- **IPv4 Flow (Time)**: The client node sends a 512-byte UDP packet to IPv4 address 10.1.2.2. Node 1 (Router) forwards it using the global IPv4 routing table, and the server returns an echo response received at .
- **IPv6 Flow (Time)**: The client sends a 512-byte UDP packet to IPv6 address 2001:2::2. The packet is routed using explicit static routes installed on Node 0 and Node 2 pointing to Node 1 (2001:1::2), confirming successful dual-stack operation.